PRE-ALGEBRA (7TH) · UNIT 4: FACTORS, FRACTIONS, AND EXPONENTS

Space Mission Packer

Use factors, fractions, exponents, and scientific notation to pack a mission to Mars.

CP2

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Mission control has chosen your team to pack the cargo bay for a crewed mission to Mars. Supplies must be split into identical kits, cargo space is measured in fractions of the bay, and distances in space are huge.

You will use factors, fractions, exponents, and scientific notation to pack the ship and plan the journey.



Crew

6 astronauts

18-month mission

Distance

Mars ≈ 2.25×10^8 km

from Earth (average)

Learning Goals

- I can use divisibility rules and the GCF to make equal groups.
- I can simplify fractions.
- I can use exponents and scientific notation for large numbers.

Project Timeline

When	What you will do
Day 1	Equal Kits
Day 2	Cargo Fractions
Day 3	Space Numbers
Final	Mission Packing Manifest + reflection

Part 1: Equal Kits

Day 1 · Use the Supply Cards.

Supply Cards

Water Pouches	Meal Packs	Oxygen Canisters
432 pouches	360 packs	96 canisters

Medical Kits	Seed Packets	Batteries
48 kits	144 packets	240 batteries

1. Is 432 divisible by 2, 3, 6, and 9? Use divisibility rules.

2. Write the prime factorization of 96 using exponents.

   Starter: *96 = 2 × 48 = 2 × 2 × 24 = …*

3. Find the GCF of 96 and 48. What does it tell you about packing kits?

4. Find the GCF of 360 and 240. How many identical kits of meals and batteries can be made? What is in each?

Part 2: Cargo Fractions

Day 2 · The cargo bay is divided into sections.

5. Water fills 18 of the 24 cargo sections. Write this as a fraction in simplest form.

6. Food fills $\frac{12}{32}$ of the freezer. Simplify.

7. Which is larger, $\frac{3}{8}$ or $\frac{2}{5}$? Explain.

8. Write $\frac{3}{4}$ as a decimal and a percent.

Part 3: Space Numbers

Day 3 · Use exponents and scientific notation.

9. Write 10^8 in standard form. How many zeros does it have?

10. Write 2.25×10^8 km in standard form.

11. The Moon is about 384,000 km away. Write this in scientific notation.

Final Product: Mission Packing Manifest

Your final product must include:

- Kit plan with the GCF
- Prime factorizations of two supply amounts
- Cargo fractions in simplest form
- Distances in scientific notation
- Your packing manifest

Reflection

R1. Why does the GCF give the greatest number of identical kits?

R2. Why do scientists use scientific notation?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Kits, cargo fractions, and space calculations use factors, fractions, and exponents correctly. Goes beyond the minimum with a creative, realistic choice.	Kits, cargo fractions, and space calculations use factors, fractions, and exponents correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Manifest includes kit contents, cargo fractions, calculations, and a mission fact sheet. Polished, organized, and visually engaging.	Manifest includes kit contents, cargo fractions, calculations, and a mission fact sheet. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.